

0.1 Monotone Functions

Theorem 0.1.1. Let $f : (a, b) \rightarrow \mathbb{R}$ be a monotonically increasing function, and let $x \in (a, b)$. Then, $f(x+)$ and $f(x-)$ exist. Specifically,

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t)$$

Moreover, if $a < x < y < b$, then $f(x+) \leq f(y-)$.

Proof. Since f is monotonically increasing, for all $t \in (a, x)$, $f(t) \leq f(x)$. Therefore, the set $\{f(t) \mid a < t < x\}$ is bounded above by $f(x)$. This implies that $\sup_{a < t < x} f(t)$ exists and satisfies $\sup_{a < t < x} f(t) \leq f(x)$. Given $\varepsilon > 0$, by the property of supremum, there exists $\delta > 0$ such that

$$\sup_{a < t < x} f(t) - \varepsilon < f(x - \delta) \leq \sup_{a < t < x} f(t).$$

Since f is increasing, for $x - \delta \leq t < x$, we have $f(x - \delta) \leq f(t) \leq \sup_{a < t < x} f(t)$, which implies

$$\sup_{a < t < x} f(t) - f(t) < \varepsilon \quad \text{for all } x - \delta \leq t < x.$$

Therefore, for any sequence $\{x_n\}$ in (a, x) , $x_n \rightarrow x$ implies $f(x_n) \rightarrow \sup_{a < t < x} f(t)$, which shows

$$\sup_{a < t < x} f(t) = f(x-).$$

Meanwhile, the second assertion $f(x+) \leq f(y-)$ for $x < y$ can be proven by observing:

$$f(x+) = \inf_{x < t < b} f(t) = \inf_{x < t < y} f(t) \quad \text{and} \quad f(y-) = \sup_{a < t < y} f(t) = \sup_{x < t < y} f(t).$$

□

Corollary 0.1.2. The set of points of discontinuity of a monotone function is at most countable.

Proof. Let $f : (a, b) \rightarrow \mathbb{R}$ be a monotone increasing function, and let E be the set of its points of discontinuity. Since f increases monotonically, $x \in E$ if and only if $f(x-) < f(x+)$.

By the density of rational numbers, for each $x \in E$, we can choose a rational $r(x) \in \mathbb{Q}$ such that

$$f(x-) < r(x) < f(x+).$$

If $x_1, x_2 \in E$ with $x_1 < x_2$, then from the previous theorem, we have:

$$r(x_1) < f(x_1+) \leq f(x_2-) < r(x_2).$$

This implies $r(x_1) < r(x_2)$, which shows that the map $r : E \rightarrow \mathbb{Q}$ is injective, thus E is at most countable. □